

TASK FLOW - TASK MANAGEMENT APP

Project Overview

TaskFlow is a lightweight task management web application. The application focuses on simplicity, usability, and clean workflow management while implementing the required core features.

The system allows users to:

- Sign in using Google Authentication
- Create and manage tasks
- Track task progress through defined workflow states
- Update task statuses

1. Application Flow

1.1 User Authentication Flow

User Opens Application

Login Screen Displayed

User Clicks "Sign in with Google"

User Selects Google Account

Firebase Auth Validates User

User Redirected to Dashboard

- Google Authentication is implemented using Firebase Authentication.
- Each user's tasks are stored separately and securely.

2. Task Management Flow

2.1 Create Task Flow

User Opens Dashboard

User Enters Task Details

User Selects Initial Status

User Clicks "Add Task"

Task Stored in Firestore Database

Task Appears in Task List

Each task contains:

- Task Title

- Task Description
- Task Status
- Creation Timestamp

2.2 Task Status Workflow

The application follows a structured workflow to maintain logical task progression.

Planned → In Progress → Completed

3. Dashboard

3.1 Main Dashboard Features

- Add new tasks
- View all tasks
- Search tasks instantly
- Filter tasks by status
- Edit tasks inline
- Delete tasks
- Update task status

4. Technology Stack

Technology	Purpose
React (Vite)	Frontend
Firebase Authentication	Google Login
Firestore Database	Database
JavaScript	Application Logic

5. Assumptions Made

Google Sign-In chosen

Tasks assumed private

Edit or Delete Tasks

Search and Filter the tasks

6. Setup Instructions

6.1 Clone Repository

git clone <repository-url>

6.2 Install Dependencies

npm install

6.3 Configure Firebase

Create a .env file in the root directory.

VITE_FIREBASE_API_KEY=your_api_key

VITE_FIREBASE_AUTH_DOMAIN=your-project.firebaseio.com

VITE_FIREBASE_PROJECT_ID=your-project-id

VITE_FIREBASE_STORAGE_BUCKET=your-project.firebaseiostorage.app

VITE_FIREBASE_MESSAGING_SENDER_ID=your_sender_id

VITE_FIREBASE_APP_ID=your_app_id

6.4 Enable Firebase Services

Enable Authentication

- Open Firebase Console
- Authentication → Sign-in Method
- Enable Google Provider

Enable Firestore Database

- Create Firestore Database
- Start in Test Mode

6.5 Run Application

npm run dev

7. Limitations

- Google-only login - No email/password authentication
- Internet required – No offline support
- No notifications - Tasks do not generate reminders

8. Important Notes for Users

- Deleted tasks cannot be recovered.
- Users must sign in with a valid Google account.

- All task changes are saved to the cloud database.

9. AI Usage Summary

AI Tools Used

- ChatGPT
- AI coding assistance tools inside IDE (Copilot)

How AI Was Used

- UI planning
- React components
- CSS refinement
- Workflow logic

Manual Improvements Made

- Firebase configuration handling
- Firestore logic corrections
- State transition validation

10. Conclusion

TaskFlow is a simple and user-friendly project-based task management application developed using AI-assisted tools. The application successfully implements Google Authentication, project management, task tracking, and status updates while maintaining a clean and minimal design.